MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR, PNP, GERMANIUM, LOW-POWER

TYPES 2N43A AND 2N44A

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for PNP, germanium, low-power transistor.

1.2 Physical dimensions. See figure 1.

1.3 Maximum ratings.

P_T 1/ T_A = 25°C	V_{CB}	V_{EB}	V_{CER}	T_J	T_{stg}
mW	Vdc	Vdc	Vdc	°C	°C
150	-45	-5	-30	+85	-65 to +100

1/ Derate linearly 2.50 mW/°C for T_A >25°C.

1.4 Primary electrical characteristics.

Limits	Type	h_{fe} V_{CE} = -5 Vdc I_E = 1.0 mAdc	h_{ib} V_{CB} = -5 Vdc I_E = 1.0 mAdc	h_{ob} V_{CB} = -5 Vdc I_E = 1.0 mAdc	f_{hfb}	C_{obo} V_{CB} = -5 Vdc I_E = 0; 100 kHz≤f≤1 MHz	NF V_{CB} = -5 Vdc; I_E = 1.0 mAdc; f = 1 kHz
			ohms	µmho	MHz	pF	dB
Min	2N43A	30	25	0.1	0.5	20	---
	2N44A	16	27	0.1	0.5	20	---
Max	2N43A	66	35	1.5	3.5	60	20
	2N44A	38	38	1.5	3.0	60	15

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

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STANDARDS

MILITARY

MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.
MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General. Requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-S-19500.

3.3 Design, construction, and physical dimensions. Transistor shall be of the design, construction, and physical dimensions shown on figure 1.

3.4 Performance characteristics. Performance characteristics shall be as specified in tables I, II, and III.

3.5 Marking. The following marking specified in MIL-S-19500 may be omitted from the body of the transistor at the option of the manufacturer:

(a) Country of origin.
(b) Manufacturer's identification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall consist of the examinations and tests specified in tables I, II, and III.

4.3 Quality conformance inspection. Quality conformance inspection shall consist of groups A, B, and C inspections.

4.3.1 Group A inspection. Group A inspection shall consist of the examinations and tests specified in table I.

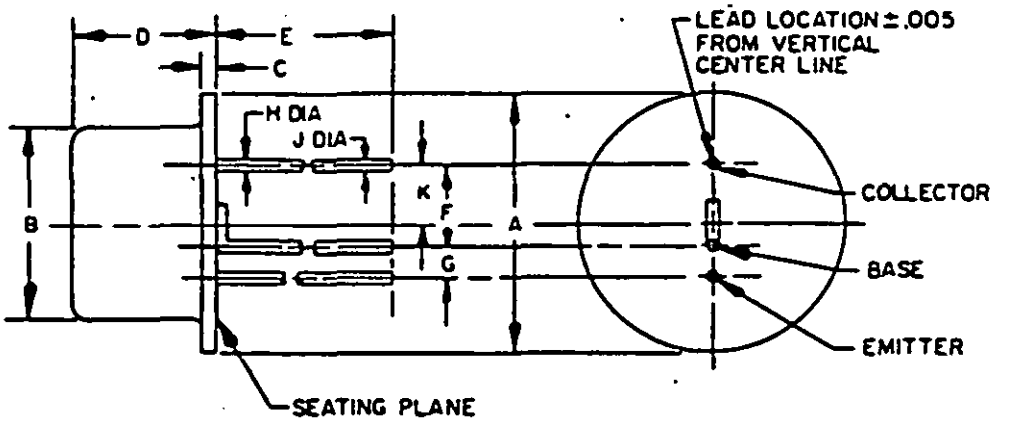
4.3.2 Group B inspection. Group B inspection shall consist of the examinations and tests specified in table II.

4.3.3 Group C inspection. Group C inspection shall consist of the examinations and tests specified in table III. This inspection shall be conducted on the initial lot and thereafter every 6 months during production.

4.3.4 Group B and group C life-test samples. Samples that have been subjected to group B, 340-hours life-test, may be continued on test for 1,000 hours in order to satisfy group C life-test requirements. These samples shall be predesignated, and shall remain subjected to the group C 1,000 hour acceptance evaluation after they have passed the group B, 340-hour acceptance criteria. The cumulative total of failures found during 340-hour test and during the subsequent interval up to 1,000 hours shall be computed for 1,000-hour acceptance criteria, see 4.3.3.

4.4 Methods of examination and test. Methods of examination and test shall be as specified in tables I, II, and III, and as follows:

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LTR	DIMENSIONS				NOTES
	INCHES		MILLIMETERS		
	MIN	MAX	MIN	MAX	
A	.430	.460	10.92	11.68	
B	.322	.345	8.18	8.76	
C	.020	.045	.51	1.14	
D	.185	.250	4.70	6.35	
E	1.500	1.750	38.10	44.45	6
F	.137	.151	3.48	3.84	
G	.041	.055	1.04	1.40	
H	.016	.019	.41	.48	3, 6
J	.016	.026	.41	.66	4, 6
K	.086	.106	2.18	2.69	

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Measured in the zone .050(1.27 mm) and .250(6.35 mm) from the seating plane.
4. Measured in the zone beyond .250(6.35 mm) from the seating plane.
5. The base shall be connected to the case.
6. All 3 leads.

FIGURE 1. Physical dimensions of transistor types 2N43A and 2N44A.

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TABLE I. Group A inspection.

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 1			10				
Visual and mechanical examination	2071			---	---	---	---
Subgroup 2			5				
Collector to base cutoff current	3036	Bias condition D: V_{CB} = -45 Vdc		I_{CBO}	---	-16	µAdc
Emitter to base cutoff current	3061	Bias condition D; V_{EB} = -5.0 Vdc		I_{EBO}	---	-10	µAdc
Breakdown voltage, collector to emitter	3011	Bias condition B: I_C = -0.6 mAdc; R_{BE} = 10 K ohms		BV_{CER}	-30	---	Vdc
Floating potential	3020	V_{CB} = -30 Vdc; voltmeter input resistance $\geq$ 1 megohm		V_{EBF}	---	-1.0	Vdc
Collector to emitter voltage (saturated)	3071			$V_{CE(sat)}$			
2N43A		I_C = -20 mAdc; I_B = -1.3 mAdc			---	-0.16	Vdc
2N44A		I_C = -20 mAdc; I_B = -2.0 mAdc			---	-0.16	Vdc
Base emitter voltage (nonsaturated)	3066	Test condition B; I_C = -20 mAdc; V_{CE} = -1.0 Vdc		V_{BE}			
2N43A					---	-0.32	Vdc
2N44A					---	-0.34	Vdc
Subgroup 3			10				
Forward-current transfer ratio	3076	V_{CE} = -1.0 Vdc; I_C = -20 mAdc		h_{FE}			
2N43A					34	65	---
2N44A					18	43	---
Forward-current transfer ratio	3076	V_{CE} = -1.0 Vdc; I_C = -100 mAdc		h_{FE}			
2N43A					30	---	---
2N44A					13	---	---
Small-signal short-circuit forward-current transfer ratio	3206	V_{CE} = -5.0 Vdc; I_E = 1.0 mAdc		h_{fe}			
2N43A					30	66	---
2N44A					16	38	---
Small-signal short-circuit input impedance	3201	V_{CB} = -5.0 Vdc; I_E = 1.0 mAdc		h_{ib}			
2N43A					25	35	ohms
2N44A					27	38	ohms

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TABLE I. Group A inspection. - Continued

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 3 - Continued							
Small-signal open-circuit output admittance	3216	V_{CB} = -5.0 Vdc; I_E = 1.0 mAdc		h_{ob}	0.1	1.5	µmho
Small-signal open-circuit reverse-voltage transfer ratio	3211	V_{CB} = -5.0 Vdc; I_E = 1.0 mAdc		h_{rb}			
2N43A					1	15	$x10^{-4}$
2N44A					1	13	$x10^{-4}$
Small-signal short-circuit forward-current transfer-ratio cutoff frequency	3301	V_{CB} = -5.0 Vdc; I_E = 1.0 mAdc		f_{hfb}			
2N43A					0.5	3.5	MHz
2N44A					0.5	3.0	MHz
Open circuit output capacitance	3236	V_{CB} = -5.0 Vdc; I_E = 0; 100 kHz $\leq f \leq$ 1 MHz		C_{obo}	20	60	pF
Noise figure	3246	V_{CB} = -5.0 Vdc; I_E = 1.0 mAdc		NF			
2N43A					---	20	dB
2N44A					---	15	dB
Subgroup 4			10				
High-temperature operation:		T_A = +70°C					
Collector to base cutoff current	3036	Bias condition D; V_{CB} = -45 Vdc		I_{CBO}	---	225	µAdc
Low-temperature operation:		T_A = -55°C					
Forward-current transfer ratio	3076	V_{CE} = -1.0 Vdc; I_C = -20 mAdc		h_{FE}			
2N43A					16	---	---
2N44A					10	---	---

TABLE II. Group B inspection.

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 1			20				
Physical dimensions	2066	(See figure 1)		---	---	---	---
Subgroup 2			15				
Solderability	2026			---	---	---	---
Thermal shock (temperature cycling)	1051	Test condition B, except step 3, T_A = +100°C		---	---	---	---

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TABLE II. Group B inspection. - Continued

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 2 - Continued							
Thermal shock (glass strain)	1056	Test condition A		---	---	---	---
Hermetic seal	1071	Test condition G or H for fine leaks; test condition A, C, D, or F for gross leaks		---	---	$5x10^{-7}$	atm cc/sec
Moisture resistance	1021			---	---	---	---
End points:							
Collector to base cutoff current	3036	Bias condition D; V_{CB} = -45 Vdc		I_{CBO}	---	-16	µAdc
Emitter to base cutoff current	3061	Bias condition D; V_{EB} = -5.0 Vdc		I_{EBO}	---	-10	µAdc
Forward-current transfer ratio	3076	V_{CE} = -1.0 Vdc; I_C = -20 mAdc		h_{FE}			
2N43A					34	65	---
2N44A					18	43	---
Subgroup 3			10				
Shock	2016	Nonoperating; 1,500 G; approx. 0.5 ms; 5 blows in each orientation: X_1, Y_1, Y_2, and Z_1		---	---	---	---
Vibration, variable frequency	2056	Nonoperating		---	---	---	---
Constant acceleration	2006	10,000 G; in each orientation: X_1, Y_1, Y_2, and Z_1		---	---	---	---
End points:							
(Same as subgroup 2)							
Subgroup 4			20				
Terminal strength (lead fatigue)	2036	Test condition E		---	---	---	---
End points:							
Hermetic seal	1071	Test condition G or H for fine leaks; test condition A, C, D, or F for gross leaks		---	---	$5x10^{-7}$	atm cc/sec
Subgroup 5			20				
Salt atmosphere (corrosion)	1041			---	---	---	---
Subgroup 6			5				
High temperature life (nonoperating)	1032	T_{stg} = 100°C; time = 340 hours (see 4.3.4)		---	---	---	---

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TABLE II. Group B inspection. - Continued

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 6 - Continued							
End points:							
Collector to base cutoff current	3036	Bias condition D; V_{CB} = -45 Vdc		I_{CBO}	---	-20	µAdc
Forward-current transfer ratio	3076	V_{CE} = -1.0 Vdc; I_C = -20 mAdc		h_{FE}			
2N43A					29	80	---
2N44A					15	60	---
Subgroup 7			5				
Intermittent operation life	1037	T_A = 25°C; P_T = 150 mW; V_{CB} = -15 Vdc; Cycles: ON, 50 ±2 minutes, OFF, 10 ±2 minutes; time = 340 hours (see 4.3.4)		---	---	---	---	---
End points:							
(Same as subgroup 6)							

TABLE III. Group C inspection.

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 1			20				
Thermal resistance	3151			θ_{J-A}	---	0.40	°C/mW
Subgroup 2			10				
Resistance to solvents	---	Method 215 of MIL-STD-202 (see 4.4.1)		---	---	---	---
Subgroup 3			λ = 10				
High-temperature life (nonoperating)	1031	T_{stg} = +100°C (see 4.3.4)		---	---	---	---
End points:							
(Same as subgroup 6 of group B inspection)							
Subgroup 4			λ = 10				
Intermittent operation life	1036	T_A = 25°C; P_T = 150 mW; V_{CB} = -15 Vdc; Cycles: ON, 50 ±2 minutes, OFF, 10 ±2 minutes (see 4.3.4)		---	---	---	---

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4.4.1 Resistance to solvents. Transistors shall be tested in accordance with method 215 of MIL-STD-202. The following details shall apply:

(a) All areas of the transistor body where marking has been applied shall be brushed.
(b) After subjection to the tests, there shall be no evidence of mechanical damage to the device and markings shall have remained legible.

5. PREPARATION FOR DELIVERY

5.1 See MIL-S-19500, section 5.

6. NOTES

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

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